ISO Standard-Based Energy Audit and 10-Year Energy Demand Forecasting for Barangay Sunvalley, Parañaque City, Philippines

4386 Daffodil St, Sun Valley, Parañaque City, 1709 Metro Manila, Philippines

In Partial Fulfillment of the Requirements for the Course
Elective 1: Renewable Energies

by

GAVINA, CLARENCE JOSEPH
HOGAT, JAN RIC
DE MESA, JOHN ROBIN
CARIASO, DRANREB DAVE D.

Submitted to
Engr. Orland D. Tubola

JANUARY 2025

TABLE OF CONTENTS

Page

Title Page …… I
Table of Contents …… II

Chapter

BACKGROUND …… 1
1

A. General Information of Audited of Audited Organization ….1

B. Occupancy and Use Schedules ……..…..2

C. Statement of Confidentiality (Limitations) …….……2

2 ENERGY PROFILE OF BARANGAY……...3

A. Energy Sources ……..……..3

B. Energy Bill Analysis ……..…..…….3

3 ENERGY DEMAND ANALYSIS ……..……5

A. Forecast Methodology ……..…..……..5

B. Forecast Results ……..……6

4 ENERGY AUDIT ……..……....8

A. Site Inspection Findings ……..……...8

B. Energy Performance Analysis ……..…….20

C. Evaluation of Improvement Opportunities……..…….. 21

D. Recommendations for Improvement of
Energy Performance ……..…… 24

E. Cost-Savings Analysis…….……... 25

6 CONCLUSION…….27

A. Summary of Findings and Recommendations……27

B. Next Steps……...…..……..….27

7 APPENDICES ……...……..28

A. References……..28

B. Documentation ……..…33

A. BACKGROUND

a) General Information of Audited of Audited Organization



***Figure 1**: Sun Valley Barangay Hall Building*

Parañaque is a highly urbanized, coastal city located in the National Capital Region (NCR) of the Philippines. The city is divided into 16 barangays, one of which is Barangay Sun Valley, situated in District 2 [8]. Surrounding Sun Valley are three other District 2 barangays: Barangay Merville to the north, Barangay San Martin De Porres to the east, and Barangay Don Bosco to the south and west. Based on the recent census, the barangay has a total population of 41,919 in the community [2].

The Barangay building is a three-story structure painted white with blue and yellow lining. It shares its location with a medical center, though the two functions are distinct and separate. For

the energy audit, only the barangay building was assessed, not the medical center. Due to security reasons, a floor plan was not available, but upon site inspection, the building features fourteen (14) rooms, a session hall, and a multi-purpose hall.

b) Occupancy and Use Schedules

The barangay hall is open from 8 AM to 5 PM on weekdays, staffed by barangay staff members and officers. Additionally, the safety and security room are operational 24 hours a day, seven days a week.

c) Statement of Confidentiality / Limitations

This energy audit was composed and examined by the specified auditors as a component of the fourth-year curriculum requirement for the Elective: Renewable Energies. The report is titled "ISO Standard-Based Energy Audit and 10-Year Energy Demand Forecasting for Barangay Sun Valley, Paranaque City, Philippines." In line with a verbal agreement with barangay officials, all information provided by them, along with data gathered by the auditors, will be exclusively used to meet the requirements of this academic project. We are committed to not disclosing any information that is not essential for completing the project. Following the analysis, the team will present their findings and propose energy efficiency recommendations to the barangay officials.

In discussing the limitations and necessary adjustments, a major challenge during the site inspection was the insufficient data collection regarding monthly bills. Although our objective was to gather 3 to 5 years' worth of monthly bill receipts, the auditors were only able to obtain 1 monthly bill for 2022, 2 monthly bills for 2023 and 2024. Consequently, the auditors used this data to construct a graph representing each month of the year, with values estimated from the available data.

A. ENERGY PROFILE OF BARANGAY

a) Energy Sources

The Manila Electric Company (MERALCO) is the largest electric distribution utility in the Philippines' private sector, serving 39 cities and 72 municipalities. Although its franchise area spans just over 9,685 km², which is only 3% of the Philippines' total land area, it delivers 55% of the nation's electricity output [7]. Among the areas it serves is Paranaque City, where Barangay Sun Valley is located. The barangay building is categorized under Meralco's General Service B

(GS-B) business customer type, which is applicable to non-residential customers with a contracted capacity between 5 kW and 39 kW [10].

b) Electric Bill Analysis

i. Monthly Electricity Consumption (kWh)

Table 1 presents key details, including the year, billing date, estimated consumption (Bill A and B), Total, Overall and Average Estimated Consumption based on the barangay's electric bills. Figure 2 complements this by visualizing the data in graphical form, illustrating trends and variations in electricity usage. The data was sourced directly from softcopies of electric bills provided by the barangay, covering the periods November to December 2022-2023 and October to November 2024.

Breakdown of Energy Consumption of Barangay Sun Valley				
Year	**Billing Date**	**Estimated Consumption (kWh)**		**Total Estimated Consumption (kWh)**
		BILL A	**BILL B**	
2022	29-01-2022	2300	1900	4200
2022	28-02-2022	2600	2000	4600
2022	29-03-2022	2700	2100	4800
2022	29-04-2022	3300	2500	5500
2022	29-05-2022	3800	2400	6200
2022	29-06-2022	4000	2600	6600
2022	29-07-2022	3700	1900	5600
2022	29-08-2022	3600	2200	5800
2022	29-09-2022	3200	1400	4600
2022	29-10-2022	3400	834	4234
2022	29-11-2022	2100	1100	3100
2022	29-12-2022	2800	1100	3900
2023	29-01-2023	2000	2700	4700
2023	28-02-2023	2200	3200	5400
2023	29-03-2023	2800	3500	6300

2023	29-04-2023	3400	3600	7000
2023	29-05-2023	3400	4300	7700
2023	29-06-2023	3400	3500	6900
2023	29-07-2023	3200	3700	6900
2023	29-08-2023	3100	3600	6700
2023	29-09-2023	2900	3200	6100
2023	29-10-2023	2800	3400	6200
2023	29-11-2023	2500	3100	5600
2023	29-12-2023	2400	3500	5900
2024	29-01-2024	2400	2800	5200
2024	29-02-2024	2100	3300	5400
2024	29-03-2024	2500	3600	6100
2024	29-04-2024	2800	3700	6500
2024	29-05-2024	3200	4400	7600
2024	29-06-2024	2800	3600	6500
2024	29-07-2024	2800	3800	6600
2024	29-08-2024	2800	3700	6500
2024	29-09-2024	2500	3300	5800
2024	29-10-2024	2500	3500	7000
2024	29-11-2024	2100	3200	5300

Table 1: *Monthly Energy Consumption of Sun Valley Barangay from January 2021 to November 2024*

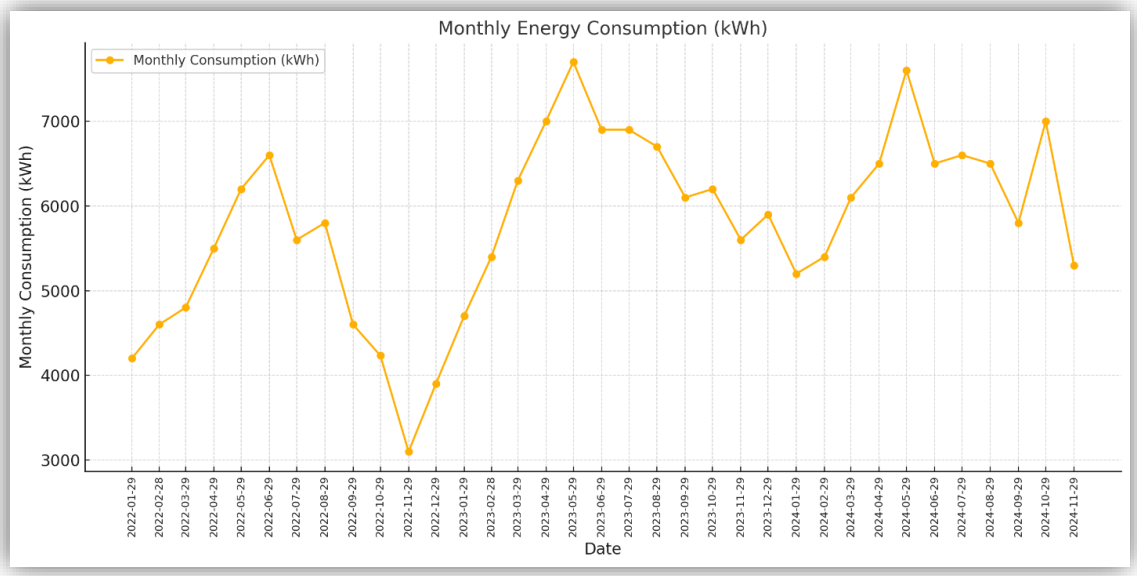


Figure 2: *Graph of Monthly Electricity Consumption*

B. ENERGY DEMAND ANALYSIS

a. Forecast Methodology

Forecasting is the process of predicting future trends or events based on historical data, enabling organizations to make informed, proactive decisions. In energy audits, accurate forecasts assist in estimating future energy demands, identifying inefficiencies, and planning sustainable resource management strategies. By leveraging forecasting, organizations can optimize resource allocation, reduce costs, and enhance sustainability efforts [4].

Prophet, developed by Facebook, is a widely used forecasting model known for its ability to handle time-series data with complex seasonal patterns, trends, and outliers. It employs an additive model framework.

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

where $y(t)$ is the forecasted value at time t. The components of this model include $g(t)$ which represents the trend capturing long-term growth or decline. $s(t)$ which accounts for periodic variations such as seasonal cycles. $h(t)$ which incorporates holiday effects or specific events influencing the data. Lastly, ϵ_t the error term accounting for unexplained variations.

This model is suitable for energy audits where consumption patterns often exhibit regular seasonal and daily variations influenced by factors like weather and operations [9]. Prophet's flexibility allows auditors to model long-term trends effectively, accommodating periodic fluctuations and unexpected changes, which is essential for generating accurate 10-year energy forecasts.

In this energy audit, the auditor used Jupyter Notebook due to its interactive and flexible environment that combines code, outputs, and documentation in a single platform. Jupyter's support for Python, the programming language used to implement Prophet, makes it a suitable choice for executing, visualizing, and iterating forecasting models. Additionally, it fosters reproducibility and collaboration, ensuring that the results are well-documented and easily shared among stakeholders [5].

b. Forecast Results

This section outlines the anticipated energy demand for the barangay over the upcoming decade. Figure 3 showcases the 10-year forecast of energy consumption, emphasizing both the underlying trends and seasonal fluctuations. Table 2 presents a comprehensive breakdown, listing each year alongside its respective peak energy demand (kWh) from 2025 to 2035.

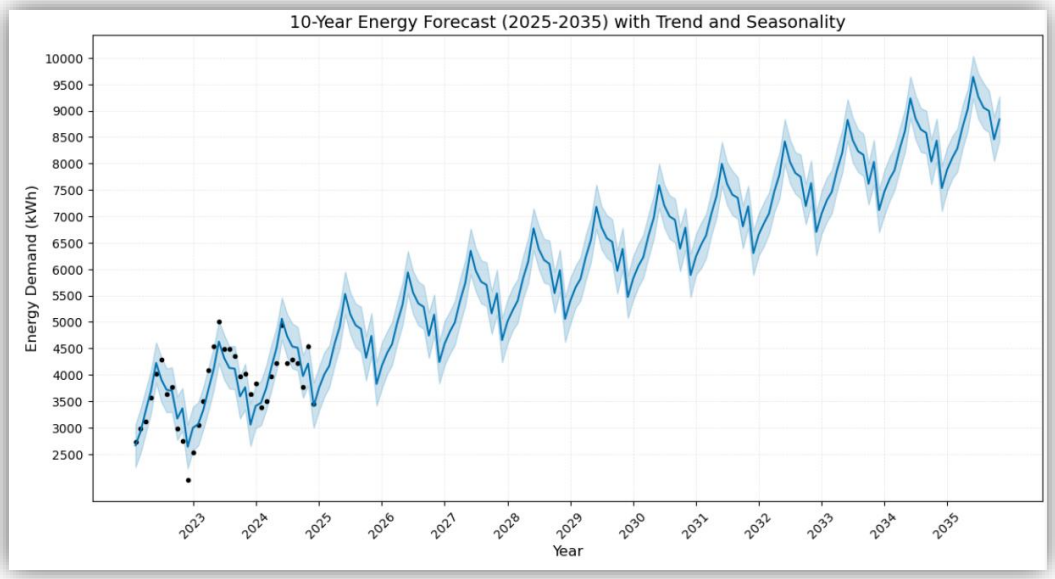


***Figure 3**: 10-Year Peak Energy Demand Forecast (2025–2035)*

YEAR	Peak Energy Demand (kWh)
2025	**5531.946**
2026	5939.736
2027	6347.249
2028	6771.009
2029	7179.079
2030	7586.869
2031	7994.382
2032	8418.142
2033	8826.212
2034	9234.002
2035	9641.515

***Table 2**: Forecasted Peak Energy Demand by Year (2025–2035)*

The forecasted energy demand from 2025 to 2035 shows a steady increase, starting at 5,531.946 kWh in 2025 and peaking at 9,641.515 kWh by 2035. This represents a significant 74% growth over the 10-year period, indicating robust and consistent energy consumption growth. The year-on-year increase ranges from 400 to 500 kWh, reflecting controlled, incremental growth in demand. Seasonal fluctuations are evident in the forecast, with demand spikes during peak periods likely tied to hotter months or significant events. However, the growth remains stable, which may reflect effective policies or infrastructure that mitigate sudden surges in demand.

C. ENERGY AUDIT

a) Site Inspection Findings

i. Lighting Systems

Breakdown of Energy Consumption by Usage and Source

Fixture/ Model	Quantity	Watt Rating (W)	Est. Hours Used per Day	Location /Room	Number of Working	Number of Not Working	Monthly Consumption (kWh)	Surveyed By:
Lighting System								
2x28-2x36 watt Fluorescent Grill Surface Light	11	28	24 hours (5 days per week)	Security Room	11	0	54.208	Robin
Pin Light	1	12	24 hours (5 days per week)	Security Room	1	0	2.112	Robin
Pin Light	7	12	8 hours (5 days per week)	TeleCPU Center	7	0	14.784	Robin
LED Light Bulb	1	8	8 hours (5 days per week)	TeleCPU Center	1	0	1.408	Robin
2x28-2x36 watt Fluorescent Grill Surface Light	11	28	8 hours (5 days per week)	Admin's Office	11	0	54.208	Robin
2x28-2x36 watt Fluorescent Grill Surface Light	8	28	8 hours (5 days per week)	Lupon Tagapamayapa's Office	8	0	39.424	Robin
2x28-2x36 watt Fluorescent Grill	4	28	8 hours (5 days per week)	Library	4	0	19.712	Robin

Surface Light								
2x28-2x36 watt Fluorescent Grill Surface Light	4	28	8 hours (5 days per week)	1st Floor Hallway/ Corridor	4	0	19.712	Robin
2x28-2x36 watt Fluorescent Grill Surface Light	5	28	8 hours (5 days per week)	Office of the Treasurer	5	0	24.64	Robin
Pin Light	5	12	8 hours (5 days per week)	Office of the Treasurer	5	0	10.56	Robin
LED Light Bulb	4	8	8 hours (5 days per week)	Office of the Treasurer	4	0	5.632	Robin
2x28-2x36 watt Fluorescent Grill Surface Light	4	28	8 hours (3 days per week)	Office of the SK Chairwoman	4	0	19.712	Robin
2x28-2x36 watt Fluorescent Grill Surface Light	7	28	8 hours (5 days per week)	Barangay Affairs Office	7	0	34.496	Robin
2x28-2x36 watt Fluorescent Grill	10	28	8 hours (5 days per week)	Office of the Secretary	10	0	49.28	Robin

Surface Light								
LED Light Bulb	1	8	8 hours (5 days per week)	Office of the Secretary	1	0	1.408	Robin
LED Light Bulb	6	8	8 hours (5 days per week)	Office of the Brgy. Captain	6	0	8.448	Robin
Pin Light	2	12	8 hours (5 days per week)	Office of the Brgy. Captain	2	0	4.224	Robin
2x28-2x36 watt Fluorescent Grill Surface Light	5	28	8 hours (5 days per week)	Kitchen/ Pantry	5	0	24.64	Robin
LED Light Bulb	1	8	8 hours (5 days per week)	Kitchen/ Pantry	1	0	1.408	Robin
2x28-2x36 watt Fluorescent Grill Surface Light	8	28	8 hours (5 days per week)	2nd Floor Hallway/ Corridor	8	0	39.424	Robin
2x28-2x36 watt Fluorescent Grill Surface Light	32	28	8 hours (2 days per week)	Multi-Purpose Hall	32	0	57.344	Robin
Pin Light	21	12	8 hours (2 days per week)	Session Hall	21	0	16.128	Robin
LED Light Bulb	1	8	8 hours (2 days per week)	Session Hall	1	0	0.512	Robin

2x28-2x36 watt Fluorescent Grill Surface Light	2	28	8 hours (5 days per week)	3rd Floor Hallway/ Corridor	2	0	9.856	Robin

ii. Cooling Equipment

Breakdown of Energy Consumption by Usage and Source								
Fixture/ Model	**Quantity**	**Watt Rating (W)**	**Est. Hours Used per Day**	**Location /Room**	**Number of Working**	**Number of Not Working**	**Monthly Consumption (kWh)**	**Surveyed By:**
AirCon and E-fan								
Kolin KSM-IW20-WCT10 M1M32 2.0 HP Split Type Air conditioner	2	900	20 hours (7 days per week)	Security Room	2	0	1980	Robin
Asahi Stand Fan 20" (508 mm) 3-leaf AS plastic fan blades	1	120	1 hour (7 days per week)	Security Room	1	0	17.4	Robin

Dowell IFST-18PR	1	75	1 hour (7 days per week)	Security Room	1	0	9	Robin
Kolin KSM-IW20-WCT10 M1M32 2.0 HP Split Type Air conditioner	1	900	8 hours (5 days per week)	TeleCPU Center	1	0	290.4	Robin
Kolin KLG-IF40-5G1M32 4.0 HP/3.0 TR Floor Mounted Air conditioner	2	1820	8 hours (5 days per week)	Admin's Office	2	0	1126.4	Robin
Kolin KSM-IW20-WCT10 M1M32 2.0 HP Split Type Air conditioner	1	900	8 hours (5 days per week)	Lupon Tagapamayapa's Office	1	0	290.4	Robin
Kolin KSM-IW20-	1	900	8 hours (5 days per week)	Library	1	0	290.4	Robin

WCT10 M1M32 2.0 HP Split Type Air conditio ner								
Kolin KSM-IW20-WCT10 M1M32 2.0 HP Split Type Air conditio ner	2	900	8 hours (5 days per week)	Office of the Treasurer	2	0	580.8	Robin
Kolin KLG-IF40-5G1M3 2 4.0 HP/3.0 TR Floor Mounte d Air conditio ner	1	1820	8 hours (5 days per week)	Office of the Treasurer	1	0	204.8	Robin
Kolin KSM-IW20-WCT10 M1M32 2.0 HP Split Type Air	1	900	8 hours (3 days per week)	Office of SK Chairwo man	1	0	158.4	Robin

conditioner								
Kolin KSM-IW20-WCT10 M1M32 2.0 HP Split Type Air conditioner	1	900	8 hours (5 days per week)	Barangay Affairs Office	1	0	290.4	Robin
Kolin KSM-IW20-WCT10 M1M32 2.0 HP Split Type Air conditioner	2	900	8 hours (5 days per week)	Office of the Secretary	2	0	580.8	Robin
Kolin KSM-IW20-WCT10 M1M32 2.0 HP Split Type Air conditioner	1	900	8 hours (5 days per week)	Office of the Brgy. Captain	1	0	290.4	Robin
Kolin KSM-IW20-WCT10 M1M32	1	900	8 hours (5 days per week)	Kitchen/ Pantry	1	0	290.4	Robin

2.0 HP Split Type Air conditioner								
Asahi Stand Fan 20” (508 mm) 3-leaf AS plastic fan blades	1	120	2 hours (5 days per week)	Kitchen/ Pantry	1	0	6.38	Robin
Kolin KLG-IF40-5G1M32 4.0 HP/3.0 TR Floor Mounted Air conditioner	2	1820	8 hours (2 days per week)	Multi-purpose Hall	2	0	409.6	Robin
Kolin KSM-IW20-WCT10M1M32 2.0 HP Split Type Air conditioner	1	900	8 hours (2 days per week)	Session Hall	1	0	105.6	Robin

Kolin KLG-IF40-5G1M3 2 4.0 HP/3.0 TR Floor Mounted Air conditioner	1	1820	8 hours (2 days per week)	Session Hall	1	0	204.8	Robin

iii. Other Systems/Auxiliary Appliances

Breakdown of Energy Consumption by Usage and Source								
Fixture/ Model	Quantity	Watt Rating (W)	Est. Hours Used per Day	Location/ Room	Number of Working	Number of Not Working	Monthly Consumption (kWh)	Surveyed By:
Auxiliary System								
PC/Desktop	1	115	8 hours (5 days per day)	Security Room	1	0	20.24	Robin
Epson Printer L360	1	13	4 hours (5 days per week)	Security Room	1	0	1.144	Robin
TV	4	50	8 hours (5 days per week)	Security Room	4	0	38.016	Robin
Water Dispenser (Camel CWD16HC-DP2)	1	905	8 hours (5 days per week)	Security Room	1	0	159.28	Robin

PC/Desktop	4	115	8 hours (5 days per week)	TeleCPU Center	4	0	80.96	Robin
Epson Printer L360	1	13	4 hours (5 days per week)	TeleCPU Center	1	0	1.144	Robin
PC/Desktop	5	115-210	8 hours (5 days per week)	Admin's Office	5	0	101.2	Robin
Epson Printer L360	2	13	4 hours (5 days per week)	Admin's Office	2	0	2.288	Robin
TV	1	30	8 hours (5 days per week)	Admin's Office	1	0	9.504	Robin
PC/Desktop	1	115	8 hours (5 days per week)	Lupon Tagapamayapa's Office	1	0	20.24	Robin
Epson Printer L360	1	13	4 hours (5 days per week)	Lupon Tagapamayapa's Office	1	0	1.144	Robin
Water Dispenser (Camel CWD16HC-DP2)	1	250	8 hours (5 days per week)	Lupon Tagapamayapa's Office	1	0	159.28	Robin
Xerox Machine (Kyocera TASKa	1	385	4 hours (5 days per week)	Library	1	0	33.8	Robin

lfa 1800)								
Refrigerator (Samsung Single Door)	1	250	24 hours (7 days per week)	Office of the Secretary	1	0	72	Robin
PC/Desktop	5	115	8 hours (5 days per week)	Office of the Secretary	5	0	101.2	Robin
Water Dispenser (Camel CWD16HC-DP2)	1	550	8 hours (5 days per week)	Office of the Secretary	1	0	159.28	Robin
Epson Printer L360	3	13	4 hours (5 days per week)	Office of the Secretary	3	0	3.432	Robin
PC/Desktop	3	115	8 hours (5 days per week)	Barangay Affairs Office	3	0	60.72	Robin
Epson Printer L360	2	13	4 hours (5 days per week)	Barangay Affairs Office	2	0	2.288	Robin
PC/Desktop	7	115-210	8 hours (5 days per week)	Office of the Treasurer	7	0	141.68	Robin
Epson Printer L360	3	13	4 hours (5 days per week)	Office of the Treasurer	3	0	3.432	Robin
Water Dispenser (Camel	1	905	8 hours (5 days per week)	Office of the Treasurer	1	0	159.28	Robin

CWD16HC-DP2)								
Rice Cooker	1	600	Half of an hour (5 days per week)	Office of the Treasurer	1	0	6.6	Robin
Microwave Oven	1	800	Half of an hour (5 days per week)	Office of the Treasurer	1	0	8.8	Robin
Refrigerator (Samsung Double Door)	1	200	24 hours (7 days per week)	Office of the Treasurer	1	0	144	Robin
Refrigerator (Mini)	1	70	24 hours (7 days per week)	Office of the Treasurer	1	0	50.4	Robin
TV	1	50	8 hours (5 days per week)	Office of the Treasurer	1	0	9.504	Robin
Refrigerator (Panasonic NR-BP7415N)	1	82	24 hours (7 days per week)	Kitchen/ Pantry	1	0	59.04	Robin
Microwave Oven	1	800	Half of an hour (5 days per week)	Kitchen/ Pantry	1	0	8.8	Robin
Rice Cooker	1	600	Half of an hour (5	Kitchen/ Pantry	1	0	6.6	Robin

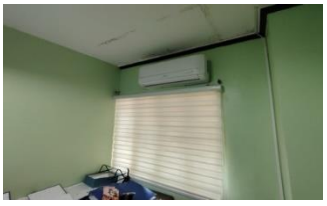
			days per week)					
TV	1	50	8 hours (2 days per week)	Session Hall	1	0	9.504	Robin

b) Energy Performance Analysis / Findings

i. Breakdown of Present Energy Consumption

Breakdown of Total Computed Energy Consumption	
Fixture	**Monthly Consumption (kWh)**
Lighting	453.504 kWh
HVAC	3908.25 kWh
Auxiliary	1124.896 kWh
Estimated Total	**5486.65 kWh**
Average kWh for 2023	**5801 kWh**
Percent Error $\frac{Actual-Expected}{Expected} x\ 100$	**5.73%**

c) Evaluation of Improvement Opportunities

Observation	Recommendation	Pictorial Evidence
Some rooms have casement windows. Some of them have cracks and gaps.	Repair or replace the damaged casement windows to prevent drafts and improve insulation.	 
Some rooms have holes and water leaks on the ceiling near the air conditioner.	Seal the holes and repair the leaks to prevent further water damage to some appliances.	 
Installed solar panels are broken and not in use.	Restore the installed solar panels to maximize and use the potential of producing renewable energy that leads to lesser electricity consumption in electricity bills.	

This rotting wood creates gaps on doors	Replace it with new materials to ensure proper sealing of doors to prevent air infiltration and improve durability.	

d) Recommendations for Improvement of Energy Performance

i. Propose specific energy-saving measures based on the gathered data:

i. Lighting Systems:

Upgrade to LED Lighting: Consider switching from fluorescent lights to LED alternatives. LEDs offer a longer lifespan, which decreases the need for frequent replacements and maintenance. They provide higher quality lighting with consistent output and no flickering or humming, enhancing visual comfort. Environmentally, LEDs are superior as they emit less heat, reduce cooling demands, and contain no mercury, simplifying disposal and minimizing hazardous waste [11]. Additionally, locations like the "Office of the Investigation" and various rooms using 2x28-2x36 watt fluorescent lights can benefit significantly from this upgrade.

By replacing the 111 fluorescent lights in Barangay Sun Valley with LED lights, the estimated monthly energy consumption is projected to decrease to 213.12 kWh. This change represents a reduction of 532.8 kWh compared to the previous consumption with fluorescent lights, underscoring the significant energy savings achieved through this upgrade.

ii. Cooling Devices:

Electric Fans that are plugged and not in use still consume electricity (known as the phantom effect), unplugging those devices if not in use can save electricity. Electric fans range up to 1.1 watts on standby power. Assuming these electric fans are left plugged in 24 hours a day, this can amount to approximately 28.512 kWh per year [1].

iii. Auxiliary Systems

Devices that are plugged in but not in use such as Televisions, Rice Cooker, and Personal Computers still consume electricity (known as the phantom effect), unplugging those devices if not in use can save electricity [3].

On average, standby power for these devices can range from 1 to 10 watts. Assuming these devices are left plugged in 24 hours a day, this can amount to approximately 87.6 kWh per year for a television, 43.8 kWh per year for a rice cooker, and 28 kWh per year for a personal computer [6].

iv. Summary of Energy Saving Recommendations

Number	Energy Saving Recommendation	Monthly Energy Saving (kWh/month)	Annual Savings	Capital Investment	Simple Payback Period
1	Replace 111 Fluorescent Tubes with LED Tubes	532.8 kWh	Php 73,526.40	Php 22,200	4 months
2	Unplugging electric fans if not in use	28.512 kWh	Php 3934.66	-	-
3	Unplugging devices if not in use like Televisions, Rice Cooker and Personal Computer	6.712	Php 926.39	-	-
TOTAL		568.024 kWh	Php 78,387.45	Php 22,200	

e) Cost-Savings Analysis

i. Formula Used

a) Monthly Energy Consumption (kWh)

Calculates the actual consumption of electricity by the building for each month.

$$MEC = \Sigma[Energy_{kW} \times t_{h/day} x\, t_{d/m}]$$

Where:

$Energy_{kw}$: Power Rating of Equipment (kW)

$t_{h/day}$: Hours of Use per Day

$t_{d/m}$: Days of Use per Month

b) Annual Energy Saving

Represents the amount of energy saved over a year as a result of implementing energy saving measures.

$$AES = \text{[Current MEC (kWh) – Estimated MEC (kWh)]}$$

where:

$Current\ MEC$: current monthly consumption

$Estimated\ MEC$: estimated monthly consumption after applying the energy saving measures

c) Annual Savings

Calculates the monetary savings due to the reduction in energy usage over one year.

$$U = 12 \times AES\left(\frac{kWh}{month}\right) \times Energy\ Rate\ \left(\frac{Php}{kWh}\right)$$

Where:

AES: Annual Energy Savings in (kWh/month)

Energy rate: Energy rate per kilowatt-hour, as charged by the energy provider

d) Simple Payback Period

Determines how long it will take for the initial cost of an energy-saving measure to be recovered, in years.

$$n_{spb} = \frac{Investment}{Annual\ Savings} = \frac{Investment}{U} = \left(\frac{\frac{Php}{Php}}{Yr}\right) = Yr$$

Where:

***Investment*:** Total Cost of proposed solution

U: Annual Savings

ii. Projected Energy Savings and Cost Reductions

Option A – Immediate		
	Average Monthly Consumption (kWh)	**Total Cost**
Current Monthly Energy Consumption	5801 kWh	Php 66,711.50
Monthly Consumption with Recommendation	5765.776 kWh	Php 66,306.42
Total Energy Saved	35.224 kWh	Php 405.08
Option B –Involve Investment		
Current Monthly Energy Consumption	5801 kWh	Php 66,711.50
Monthly Consumption with Recommendation	5232.976 kWh	Php 60,179.224
Total Energy Saved	568.104 kWh	Php 6,533.20

D. CONCLUSION

a. Summary of Findings and Recommendations

i. Key takeaways from the audit

- The cooling system significantly contributes to the building's monthly energy consumption.
- Tinted Windows can significantly reduce heat absorption.
- Maintaining the building effectively is crucial for improving the energy efficiency of appliances and lighting systems within the facility.
- Unplugging the unused appliances can significantly reduce power energy consumption

ii. Emphasis on critical recommendations and their impacts

- Since all rooms are air-conditioned, maintaining quality, operating times, and upkeep of the cooling devices has the greatest impact on reducing energy consumption and monthly costs.

b. Next Steps

i. Suggested actions for the organization

- Considering the time of operation, user manual and its maintenance for energy efficiency of cooling systems should be monitored and evaluated. Prioritize having tinted windows in each room to reduce building heat absorption.

ii. Monitoring and verification plan for implemented measures

- Implement a structured maintenance schedule for appliances with high energy consumption, such as especially the split type and floor stand air conditioners operated in each room.

E. REFERENCES

[1] A. Bevan, "How Much Does a Fan Cost to Run? 5 Tips To Save Energy (March 2024 update)," *Morris Direct*, Jul. 16, 2024. https://www.morrisdirect.co.uk/blogs/advice-centre-fans/how-much-does-a-fan-cost-to-run

[2] "COMMUNITY PROFILE," *integratedprojectji*. https://integratedproject15.wixsite.com/integratedprojectji/community-profile

[3] "Does Unplugging Appliances Save Money and Electricity? | Inspire Clean Energy," *www.inspirecleanenergy.com*. https://www.inspirecleanenergy.com/blog/sustainable-living/does-unplugging-appliances-save-electricity

[4] "Forecasting: Principles and Practice (3rd ed)," *Otexts.com*, 2021. https://otexts.com/fpp3

[5] "Forecasting: principles and practice: Hyndman, Rob J., author: Free Download, Borrow, and Streaming: Internet Archive," *Internet Archive*, 2022. https://archive.org/details/forecastingprinc0000hynd/page/8/mode/2up?

[6] G. M, "Energy Vampires: How Devices Drain Power When Not in Use," *Consumer Energy Center*, Jan. 02, 2025. https://www.consumerenergycenter.org/energy-vampire

[7] Meralco, "Corporate profile," *Meralco*, 2018. https://company.meralco.com.ph/corporate-profile

[8] PacLights, “Benefits of Retrofitting Fluorescent Lights to LED Lighting Fixtures,” *PacLights*, Jul. 30, 2024. https://www.paclights.com/learning-center/benefits-of-conversion-from-fluorescent-lights-to-led/

[9] PhilAtlas, “Parañaque City Profile – PhilAtlas,” *www.philatlas.com*. https://www.philatlas.com/luzon/ncr/paranaque.html

[10] S. J. Taylor and B. Letham, “Forecasting at Scale,” *The American Statistician*, vol. 72, no. 1, pp. 37–45, Sep. 2017, doi: https://doi.org/10.1080/00031305.2017.1380080.

[11] “Start or Modify Service,” *Meralco*. https://company.meralco.com.ph/faq-about/start-or-modify-service

F. APPENDICES

a) Code

```
# !pip install prophet

# Import necessary libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from prophet import Prophet

# Load the dataset
df = pd.read_csv('Brgy-Sunvalley-Monthly-Consumption.csv')

# Define your holidays as lists
holiday_dates = [
'2025-01-01', '2025-01-29', '2025-04-17', '2025-04-18', '2025-04-19', '2025-04-09',
'2025-05-01', '2025-06-12', '2025-08-21', '2025-08-25', '2025-11-01', '2025-11-30',
'2025-12-08', '2025-12-25', '2025-12-30', '2025-12-31', '2025-03-30', '2025-06-06',
'2025-07-27', '2025-10-31', '2025-12-24', '2026-01-01', '2026-02-17', '2026-04-02',
'2026-04-03', '2026-04-04', '2026-04-09', '2026-05-01', '2026-06-12', '2026-08-21',
'2026-08-31', '2026-11-01', '2026-11-30', '2026-12-08', '2026-12-25', '2026-12-30',
'2026-12-31', '2026-03-20', '2026-05-27', '2027-01-01', '2027-02-06', '2027-03-25',
'2027-03-26', '2027-03-27', '2027-04-09', '2027-05-01', '2027-06-12', '2027-08-21',
'2027-08-30', '2027-11-01', '2027-11-30', '2027-12-08', '2027-12-25', '2027-12-30',
'2027-12-31', '2027-03-09', '2027-05-16', '2028-01-01', '2028-01-26', '2028-04-13',
'2028-04-14', '2028-04-15', '2028-04-09', '2028-05-01', '2028-06-12', '2028-08-21',
'2028-08-28', '2028-11-01', '2028-11-30', '2028-12-08', '2028-12-25', '2028-12-30',
'2028-12-31', '2028-02-26', '2028-05-05', '2029-01-01', '2029-02-13', '2029-03-29',
'2029-03-30', '2029-03-31', '2029-04-09', '2029-05-01', '2029-06-12', '2029-08-21',
'2029-08-27', '2029-11-01', '2029-11-30', '2029-12-08', '2029-12-25', '2029-12-30',
```

```
'2029-12-31', '2029-02-14', '2029-04-24', '2030-01-01', '2030-02-03', '2030-04-18',
'2030-04-19', '2030-04-20', '2030-04-09', '2030-05-01', '2030-06-12', '2030-08-21',
'2030-08-26', '2030-11-01', '2030-11-30', '2030-12-08', '2030-12-25', '2030-12-30',
'2030-12-31', '2030-02-04', '2030-04-13', '2031-01-01', '2031-01-23', '2031-04-10',
'2031-04-11', '2031-04-12', '2031-04-09', '2031-05-01', '2031-06-12', '2031-08-21',
'2031-08-25', '2031-11-01', '2031-11-30', '2031-12-08', '2031-12-25', '2031-12-30',
'2031-12-31', '2031-01-24', '2031-04-02', '2032-01-01', '2032-02-11', '2032-03-25',
'2032-03-26', '2032-03-27', '2032-04-09', '2032-05-01', '2032-06-12', '2032-08-21',
'2032-08-30', '2032-11-01', '2032-11-30', '2032-12-08', '2032-12-25', '2032-12-30',
'2032-12-31', '2032-02-14', '2032-03-22', '2033-01-01', '2033-01-31', '2033-04-14',
'2033-04-15', '2033-04-16', '2033-04-09', '2033-05-01', '2033-06-12', '2033-08-21',
'2033-08-29', '2033-11-01', '2033-11-30', '2033-12-08', '2033-12-25', '2033-12-30',
'2033-12-31', '2033-01-02', '2033-12-23', '2033-03-11', '2034-01-01', '2034-02-19',
'2034-04-06', '2034-04-07', '2034-04-08', '2034-04-09', '2034-05-01', '2034-06-12',
'2034-08-21', '2034-08-28', '2034-11-01', '2034-11-30', '2034-12-08', '2034-12-25',
'2034-12-30', '2034-12-31', '2034-12-12', '2034-03-01'
]

holidays_df = pd.DataFrame({
    'holiday': ['PH Holiday'] * len(holiday_dates),
    'ds': pd.to_datetime(holiday_dates)
})

# Function to align holidays to the first of each month
def align_holidays_to_month(holidays_df):
    holidays_df['ds'] = holidays_df['ds'].apply(lambda x: x.replace(day=1))
    return holidays_df.drop_duplicates() # Drop duplicates if multiple holidays fall on the same month

# Apply the function to align holidays
holidays_df = align_holidays_to_month(holidays_df)

# Display the aligned holidays
print(holidays_df)

# Scale the historical data to keep the forecast
if 'y' in df.columns:
    max_value = df['y'].max()
    scaling_factor = 5000 / max_value  # Scale to a maximum of ~9,000 to allow for growth
    df['y'] = df['y'] * scaling_factor # Apply scaling

# Initialize the Prophet model
m = Prophet(
    holidays=holidays_df,
    seasonality_mode='additive',
    changepoint_prior_scale=0.01,
    seasonality_prior_scale=0.1,
    holidays_prior_scale=10,
```

```
    yearly_seasonality=True
)

# Prepare the dataset for Prophet
df['ds'] = pd.to_datetime(df['ds']) # Ensure the date column is in datetime format
m.fit(df)

# Generate future dataframe
future = m.make_future_dataframe(periods=132, freq='MS')  # Monthly forecast
forecast = m.predict(future)

# Display forecast tail
print(forecast[['ds', 'yhat', 'yhat_lower', 'yhat_upper']].tail())

# Plot the forecast
fig1 = m.plot(forecast)
fig1.set_size_inches(12, 6)
x_ticks = pd.date_range(start=forecast['ds'].min(), end=forecast['ds'].max(), freq='YS')  # Year start
plt.xticks(
    ticks=x_ticks,
    labels=[tick.strftime('%Y') for tick in x_ticks],
    rotation=45,
    fontsize=10
)
plt.xlabel('Year', fontsize=12)
y_min = forecast['yhat'].min() # Minimum forecasted value
y_max = forecast['yhat'].max() # Maximum forecasted value
y_ticks = np.arange(
    start=500 * np.floor(y_min / 500),
    stop=500 * np.ceil(y_max / 500) + 500,
    step=500
)
plt.yticks(ticks=y_ticks, fontsize=10)
plt.ylabel('Energy Demand (kWh)', fontsize=12)
plt.grid(True, which='both', linestyle='--', linewidth=0.5)
plt.title('10-Year Energy Forecast (2025-2035) with Trend and Seasonality', fontsize=14)
plt.savefig('10-Year_Energy_Forecast_2025-2035.jpg', format="jpg", dpi=300, bbox_inches="tight")
plt.show()

# Plot the input data
plt.figure(figsize=(10, 6))
plt.plot(df['ds'], df['y'], label='Monthly Energy Consumption (2022-2024)', color='blue')
plt.title('Monthly Energy Consumption (2022-2024)')
plt.xlabel('Date')
plt.ylabel('kWh')
plt.xticks(rotation=90)
plt.legend()
```

```
plt.savefig("Monthly_Energy_Consumption_2022-2024.jpg", format="jpg", dpi=300, bbox_inches="tight")
plt.show()

# Plot the components of the forecast
fig2 = m.plot_components(forecast)
fig2.savefig("forecast_components.jpg", format="jpg", dpi=300, bbox_inches="tight")

# Create a table showing the peak kWh per year for the forecasted data (2025-2035)
forecast['year'] = forecast['ds'].dt.year
peak_kwh_per_year = forecast.groupby('year')['yhat'].max() # Get peak values (max) for each year
peak_kwh_table = peak_kwh_per_year.reset_index()

# Display the peak kWh table
print("Peak kWh per year (2025-2035):")
print(peak_kwh_table)

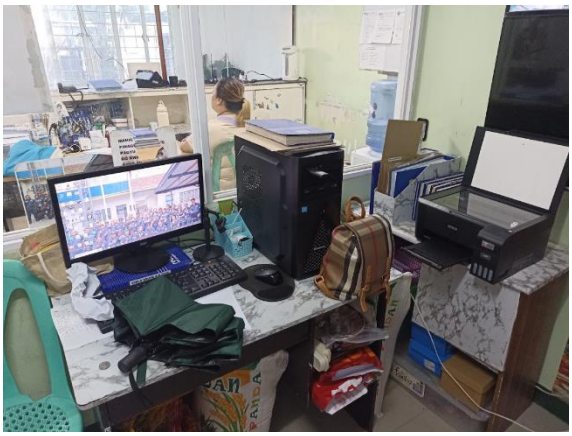
# Save the peak kWh table to a CSV file
peak_kwh_table.to_csv("peak_kwh_output.csv", index=False)
```

b) Documentation

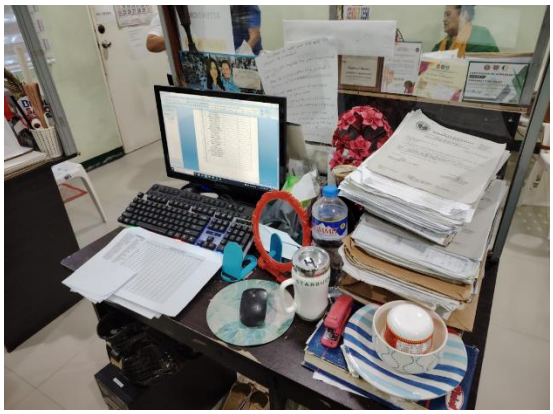
- **Security Room**

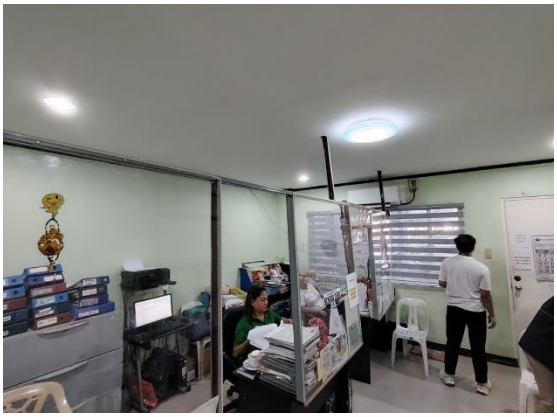


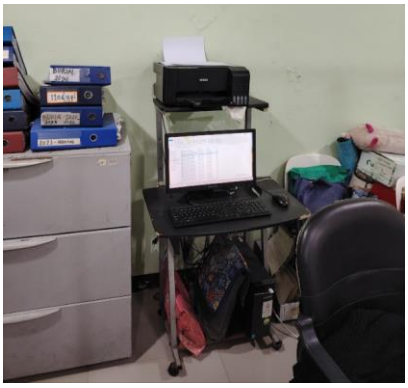




- **TeleCPU Center**







- **Library**







- **Admin's Office**







- **Lupon Tagapamayapa's Office**

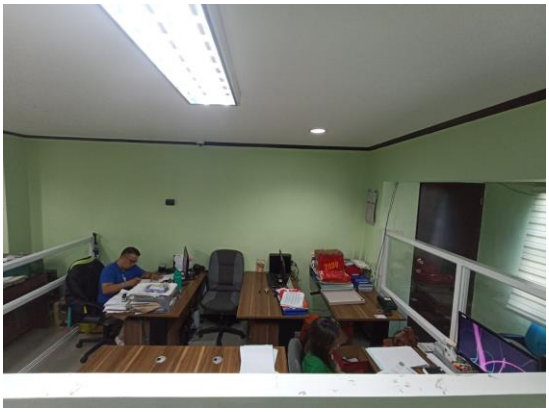






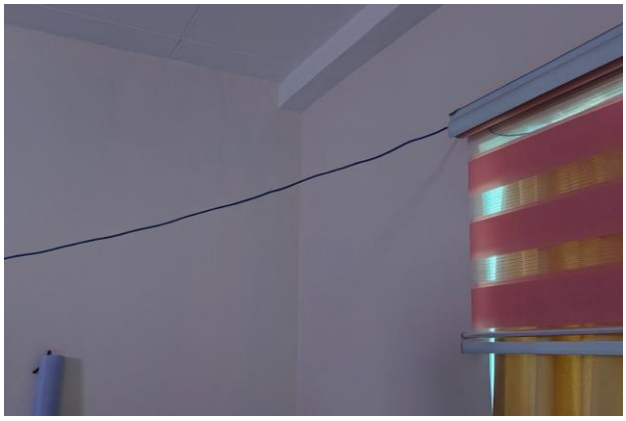
- **Office of the Treasurer**







- **Office of the SK Chairwoman**





- **Barangay Affairs Office**

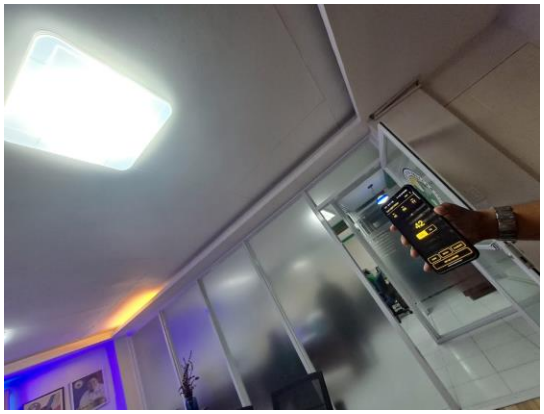






- **Office of the Secretary**



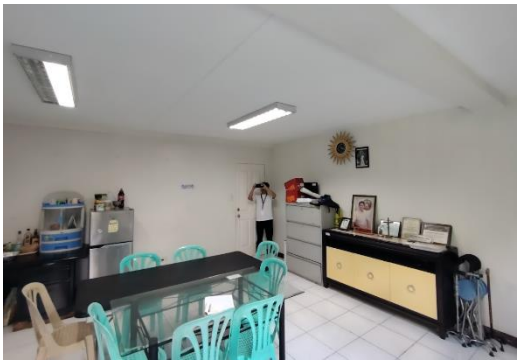




- **Kitchen/Pantry**





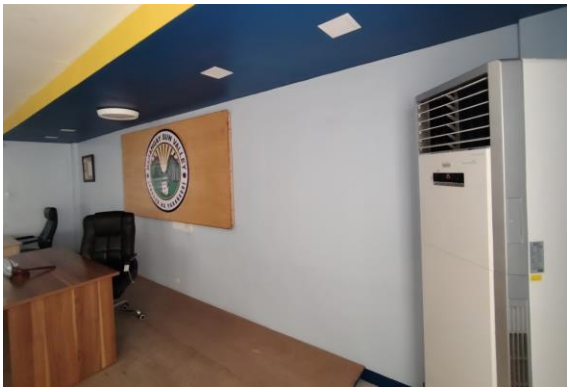


- **Multi-Purpose Hall**





- **Session Hall**


c) **Monthly bill**